

New Mutations and Mutagens

Mutations can arise in a number of ways. Errors during DNA replication or recombination can lead to nucleotide-pair substitutions, insertions, or deletions, as well as to mutations affecting longer stretches of DNA. If an incorrect nucleotide is added to a growing chain during replication, for example, the base on that nucleotide will then be mismatched with the nucleotide base on the other strand. In many cases, the error will be corrected by DNA proofreading and repair systems (see Concept 16.2). Otherwise, the incorrect base will be used as a template in the next round of replication, resulting in a mutation. Such mutations are called *spontaneous mutations*. It is difficult to calculate the rate at which such mutations occur. Rough estimates have been made of the rate of mutation during DNA replication for both *E. coli* and eukaryotes, and the numbers are similar: About one nucleotide in every 10^{10} is altered, and the change is passed on to the next generation of cells.

A number of physical and chemical agents, called **mutagens**, interact with DNA in ways that cause mutations. In the 1920s, Hermann Muller discovered that X-rays caused genetic changes in fruit flies, and he used X-rays to make *Drosophila* mutants for his genetic studies (just as Beadle and Tatum did with their *Neurospora* cells). But Muller also recognized an alarming implication of his discovery: X-rays and other forms of high-energy radiation pose hazards to the genetic material of people as well as laboratory organisms. Mutagenic radiation, a physical mutagen, includes ultraviolet (UV) light, which can cause disruptive thymine dimers in DNA (see Figure 16.20).

Chemical mutagens fall into several categories. Nucleotide analogs are chemicals similar to normal DNA nucleotides but that pair incorrectly during DNA replication. Other chemical mutagens interfere with correct DNA replication by inserting themselves into the DNA and distorting the double helix. Still other mutagens cause chemical changes in bases that change their pairing properties.

Researchers have developed a variety of methods to test the mutagenic activity of chemicals. A major application of these tests is the preliminary screening of chemicals to identify those that may cause cancer. This approach makes sense because most carcinogens (cancer-causing chemicals) are mutagenic, and conversely, most mutagens are carcinogenic.

Using CRISPR to Edit Genes and Correct Disease-Causing Mutations

Ever since biologists have understood how disease-causing proteins were the result of mutations in genes, they have sought techniques for **gene editing**—altering genes in a specific, predictable way. Their aim has been to use such a

technique to change specific genes in living cells, in part to try to correct genes that cause disease. Over the past 15 years, biologists have developed a powerful new technique for gene editing, called the **CRISPR-Cas9 system**, that is transforming the field of *genetic engineering*, the direct manipulation of genes for practical purposes. Cas9 is a bacterial protein that helps defend bacteria against the viruses that infect them (bacteriophages). In bacterial cells, Cas9 acts together with “guide RNA” made from the CRISPR region of the bacterial genome. (Figure 19.7 explains how this defense system works in bacteria.)

➔ **Mastering Biology HHMI Video: Jennifer Doudna: CRISPR Provides a Tool for Changing Genes**



➔ **Mastering Biology Interview with Francisco Mojica: Discovering CRISPR in bacteria (see the interview before Chapter 13)**



Similar to other DNA-cutting enzymes involved in DNA repair (described in Concept 16.2), Cas9 is a nuclease that cuts double-stranded DNA molecules. The power of Cas9 for gene editing is that the Cas9 protein will cut any sequence to which it is directed. Cas9 is directed to its target by a guide RNA molecule that it binds and uses as a homing device, cutting both strands of any DNA sequence that is complementary to the guide RNA. Scientists have been able to exploit the function of Cas9 by introducing a Cas9–guide RNA complex into a cell they wish to alter (Figure 17.28). The guide RNA in the complex is engineered to be complementary to the “target” gene. Cas9 cuts both strands of the target DNA, and the resulting broken ends of DNA trigger a DNA repair system (similar to that shown in Figure 16.20). When there is no undamaged DNA for the enzymes of the repair system to use as a template, as shown at the bottom left of Figure 17.28, the repair enzymes introduce or remove random nucleotides while rejoining the ends. Generally, this process alters the DNA sequence so that the gene no longer works properly. This technique is a highly successful way for researchers to “knock out” (disable) a given gene to study what that gene does in an organism.

But what about using this system to help treat genetic diseases? Researchers have modified the technique so that the CRISPR-Cas9 system can be used to repair a gene that has a harmful mutation. They introduce a segment from the normal (functional) gene along with the CRISPR-Cas9 system. After Cas9 cuts the target DNA, repair enzymes can use the normal DNA as a template to repair the target DNA at the break point. In this way, the CRISPR-Cas9 system edits the defective gene so that it is corrected (see the bottom right of Figure 17.28).

In 2018, researchers reported promising results using the CRISPR-Cas 9 system in an attempt to correct the genetic